Chemistry Topic 9 — Discoveries, Scientists and Miscellaneous Chemistry

Covers syllabus (click to expand)

Foundational Chemists & Historical Discoveries (Lavoisier, Dalton, Priestley, Cavendish, Wöhler, Kekulé, Berzelius, Pauling) | Indian Chemistry Contributions (P. C. Ray Father of Indian Chemistry & Mercurous Nitrite, S. S. Bhatnagar, C. N. R. Rao, Asima Chatterjee) | Water Chemistry: Hard vs Soft Water (Temporary vs Permanent Hardness Removal, Clark's, Calgon, Zeolite, Permutit) | Heavy Water (D_2O) & Superheavy Water (T_2O) in Nuclear Reactors | Master Trade Names & Common Chemicals Dictionary (Lunar Caustic, Sindoor, Minium, Calomel, Corrosive Sublimate, Eosin, Vitriols, Saltpetres) | Gas Nicknames & Laboratory Tests | Landmark Nobel Prizes in Chemistry (2020–2025: AlphaFold, Quantum Dots, Click Chemistry, Organocatalysis, CRISPR)

Sources baked in: NCERT Class 9–12 (History of Chemistry; Hydrogen; Chemistry in Everyday Life), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring match questions on Blue Vitriol as fungicide, Eosin as red ink, Silver Iodide in artificial rain, Zinc Phosphide as rodenticide, Lunar Caustic in voting ink, P. C. Ray and mercurous nitrite, Heavy water as nuclear moderator, and temporary vs permanent water hardness. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2024–2025	**Nobel Prize in Chemistry (2024)**	Awarded to **David Baker** (computational protein design) and **Demis Hassabis & John M. Jumper** (AI protein structure prediction via AlphaFold2), decoding the 3D structures of virtually all 200 million identified biological proteins.
2024	**Election Chemistry — Voter's Indelible Ink**	In the 2024 Indian General Elections, Mysore Paints & Varnish Ltd. supplied indelible voter ink containing **Silver Nitrate (AgNO_3, Lunar Caustic)**. When applied to the fingernail, AgNO_3 reacts with skin proteins and ambient sunlight to photochemically precipitate insoluble black metallic silver (Ag), which cannot be washed off with soap or organic solvents.
2023	**UPPCS Prelims 2023**	Questions on Gas Nicknames & Element Groups: **Laughing gas** confirmed as **Nitrous Oxide (N_2O)**. Identified that **Actinium (Ac, Z=89)** is an actinide transition element, NOT a noble gas (Group 18).
2023	**Nobel Prize in Chemistry (2023)**	Awarded to **Moungi Bawendi, Louis Brus, and Alexei Ekimov** for the discovery and synthesis of **Quantum Dots** (semiconductor nanocrystals whose optical properties are governed by quantum size confinement, widely used in QLED televisions and bio-imaging).
2020	**UPPCS Prelims 2020**	Classic Master Match Set: Blue vitriol (fungicide), Eosin (red ink), Silver iodide (artificial rain), and Zinc phosphide (rodenticide).

📋 **Master Reference Tables (Click to Expand)**

Table 1: Master Dictionary of Chemical Common / Trade Names | Trade / Common Name | Systematic IUPAC Chemical Name | Precise Chemical Formula | Primary Real-World Application / Must-Score Signature | |---|---|---|---| | **Lunar Caustic** | Silver Nitrate | **$AgNO_3$** | Indelible voting mark ink; cauterizing warts | | **Sindoor / Minium / Red Lead**| Trilead Tetroxide | **Pb_3O_4** | Traditional cosmetic vermilion; anti-rust primer | | **Litharge** | Lead Monoxide | **PbO** | Lead glass manufacture; assayer's flux | | **Quicksilver** | Elemental Mercury | **Hg** | Liquid metal; barometers, clinical thermometers | | **Calomel** | Mercurous Chloride | **Hg_2Cl_2** | Secondary reference electrode; historic medicine | | **Corrosive Sublimate** | Mercuric Chloride | **$HgCl_2$** | Powerful disinfectant; wood preserving poison | | **Vermilion** | Mercuric Sulphide | **HgS** | Red pigment; primary mercury ore (Cinnabar) | | **Eosin** | Tetrabromofluorescein disodium | $C_{20}H_6Br_4Na_2O_5$ | Characteristic red coloring dye in commercial **Red Ink** | | **Dry Ice** | Solid Carbon Dioxide | **$CO_2\text{ (solid)}$** | Sublimes at -78.5°C; cold chain food refrigerant | | **Pearl Ash** | Potassium Carbonate | **K_2CO_3** | Manufacture of optical and Bohemian glass, soaps | | **Saltpetre (Nitre)** | Potassium Nitrate | **KNO_3** | Gunpowder propellant oxidizer; fertilizers | | **Chile Saltpetre** | Sodium Nitrate | **$NaNO_3$** | Nitrogenous fertilizer; nitric acid manufacture | | **Sal Ammoniac** | Ammonium Chloride | **NH_4Cl** | Dry cell battery electrolyte; soldering metal flux | | **Laughing Gas** | Nitrous Oxide | **N_2O** | Mild inhalation anesthetic in dental surgery | | **Philosopher's Wool** | Zinc Oxide | **ZnO** | White paint pigment; calamine sunscreen ointment | | **Glauber's Salt** | Sodium Sulphate Decahydrate | **$Na_2SO_4 \cdot 10H_2O$** | Medicinal saline purgative laxative | | **Hypo Solution** | Sodium Thiosulphate Pentahydrate| **$Na_2S_2O_3 \cdot 5H_2O$** | Fixer in photography; antichlor in textile bleaching | | **Barium Meal** | Barium Sulphate | **$BaSO_4$** | Radiopaque X-ray diagnostic contrast agent | | **Blue Vitriol** | Copper Sulphate Pentahydrate | **$CuSO_4 \cdot 5H_2O$** | Bordeaux mixture fungicide; electroplating | | **Green Vitriol** | Ferrous Sulphate Heptahydrate | **$FeSO_4 \cdot 7H_2O$** | Iron nutritional deficiency tablets; dye mordant | | **White Vitriol** | Zinc Sulphate Heptahydrate | **$ZnSO_4 \cdot 7H_2O$** | Manufacture of lithopone; eye drops | | **Oil of Vitriol** | Concentrated Sulphuric Acid | **H_2SO_4** | "King of Chemicals"; lead-acid car batteries | --- ### Table 2: Water Hardness Treatment Matrix (NCERT / Lucent Standard) | Method Name | Category of Hardness Removed | Reagent / Material Employed | Chemical Mechanism / Balanced Reaction | Limitations / Notes | |---|---|---|---|---| | **Simple Boiling** | Temporary Hardness Only | Heat energy (Δ) | $Ca(HCO_3)_2 \xrightarrow{\Delta} CaCO_3\downarrow + H_2O + CO_2$ | Ineffective against sulphates/chlorides. | | **Clark's Method** | Temporary Hardness Only | Slaked Lime ($Ca(OH)_2$) | $Ca(HCO_3)_2 + Ca(OH)_2 \to 2CaCO_3\downarrow + 2H_2O$ | Excess lime causes re-hardening. | | **Washing Soda Method** | Permanent & Temporary | Sodium Carbonate (Na_2CO_3) | $CaCl_2 + Na_2CO_3 \to CaCO_3\downarrow + 2NaCl$ | Precipitator method; leaves soluble NaCl. | | **Calgon Process** | Permanent & Temporary | Sodium Hexametaphosphate ($Na_6P_6O_{18}$) | $2Ca^{2+} + [Na_4P_6O_{18}]^{2-} \to [Ca_2P_6O_{18}]^{2-} + 4Na^+$ | Sequestration; no precipitate formed. | | **Permutit / Zeolite Process** | Permanent & Temporary | Sodium Aluminium Silicate ($Na_2\text{Ze}$) | $Ca^{2+} + Na_2\text{Ze} \to Ca\text{Ze} + 2Na^+$ | Regenerated with 10% brine (NaCl). | | **Synthetic Resins** | Complete Demineralization | Cation (R-H) & Anion (R-OH) resins | $2R\text{-}H + Ca^{2+} \to R_2Ca + 2H^+$; $R\text{-}OH + Cl^- \to R\text{-}Cl + OH^-$ | Produces ultra-pure deionized water. |

Detailed Theory Notes

9.1 Landmark Chemists & Historical Paradigms

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Antoine Lavoisier  ——> Father of Modern Chemistry; Conservation of Mass; named Oxygen & Hydrog
John Dalton        ——> Atomic Theory (1808); Law of Multiple Proportions; Partial Pressures
Friedrich Wöhler   ——> Synthesized Urea from Ammonium Cyanate (1828); killed Vital Force Theor
August Kekulé      ——> Benzene Ring Structure (1865); Tetravalency of Carbon
P. C. Ray          ——> Father of Indian Chemistry; Mercurous Nitrite; Bengal Chemicals (1901)
Linus Pauling      ——> Hybridization; Electronegativity Scale; Alpha-helix Protein Structure
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1. Wöhler's Demolition of the Vital Force Theory

Until 1828, chemistry was divided by Berzelius's dogma that organic compounds possessed an inscrutable "vital force" (*vis vitalis*) and could only be synthesized within living biological organisms. - Friedrich Wöhler heated inorganic **Ammonium Cyanate (NH_4CNO)** in an attempt to prepare ammonium cyanate crystals. - Instead, the atoms spontaneously rearranged via an intramolecular isomeric shift: $NH_4CNO\text{ (Inorganic Salt)} \xrightarrow{\Delta} O=C(NH_2)_2\text{ (Urea, Organic Compound)}$ - This breakthrough demonstrated that organic chemistry is governed by the identical laws of physics and chemical bonding that govern inorganic substances.

2. Acharya P. C. Ray & the Nitrite Breakthrough

- In 1896, working in his laboratory at Presidency College, Calcutta, P. C. Ray investigated yellow crystalline deposits formed during the reaction of dilute nitric acid with mercury.
- He isolated and proved the existence of **Mercurous Nitrite ($Hg_2(NO_2)_2$)**, a compound previously deemed impossible due to the instability of nitrites of low-valent heavy metals.
- He synthesized over a dozen new stable nitrite complexes, earning recognition from the Royal Society as the **"Master of Nitrites"**.

9.2 Water Hardness & Nuclear Heavy Water Physics

1. Mechanics of Hard Water

- Rainwater naturally absorbs atmospheric CO_2 to form weak carbonic acid (H_2CO_3).
- As this acidic water percolates through limestone ($CaCO_3$) and dolomite ($CaCO_3 \cdot MgCO_3$) strata, it dissolves them into soluble bicarbonates: $CaCO_3\text{ (s)} + H_2O\text{ (l)} + CO_2\text{ (g)} \to Ca(HCO_3)_2\text{ (aq)}$
- In industrial steam boilers, hard water precipitates a thick, crusty, insulating deposit of calcium carbonate and sulphate called **Boiler Scale**, which impedes heat transfer, overheats boiler metal, and causes catastrophic pressure explosions.

2. Nuclear Heavy Water (D_2O)

- The nucleus of Protium (^1H) consists of 1 proton and 0 neutrons. The nucleus of Deuterium (^2H) consists of 1 proton and 1 neutron.

- In a nuclear reactor, fast fission neutrons collide elastically with moderator nuclei. Because the mass of a Deuterium nucleus (2\text{ u}) is comparable to a neutron (1\text{ u}), neutrons transfer a significant fraction of their kinetic energy per collision, slowing from 20,000\text{ km/s} (2\text{ MeV}) to 2.2\text{ km/s} (0.025\text{ eV}) within approximately 25 collisions.
- Crucially, Deuterium rarely absorbs neutrons to form Tritium (\sigma_a = 0.0005\text{ barns}), allowing Indian PHWRs to sustain criticality using natural unenriched uranium (0.71\% {}^{235}U) without expensive isotope enrichment.

9.3 Specialty Trade Chemicals & Diagnostic Reactions

1. Indelible Voting Ink (Lunar Caustic)

- Formulated by the National Physical Laboratory (NPL, New Delhi) and manufactured exclusively by Mysore Paints and Varnish Ltd.
- Contains **Silver Nitrate (AgNO_3, typically 10–18\%)**, mixed with a purple dye and volatile solvent.
- When brushed on the cuticle and nail, AgNO_3 permeates the dead protein layer (keratin). Upon exposure to ambient light, the silver ions are photoreduced: Ag^+ + e^- \xrightarrow{h\nu} Ag\text{ (metallic silver colloid)} The black metallic silver stain binds permanently to the protein matrix and can only disappear as the nail grows out over 2–4 weeks.

2. Barium Meal in Gastrointestinal Radiology

- Soluble barium salts (BaCl_2, Ba(NO_3)_2) are virulent cardiovascular and neuromuscular poisons (Ba^{2+} blocks potassium channels).
- However, **Barium Sulphate (BaSO_4)** has an extremely small solubility product (K_{sp} \approx 1.1 \times 10^{-10}\text{ mol}^2/\text{L}^2), releasing virtually zero free Ba^{2+} ions in stomach acid.
- Because Barium (Z=56) strongly absorbs X-ray photons via the photoelectric effect, a swallowed suspension of BaSO_4 coats the mucosal lining of the esophagus, stomach, and duodenum, casting a sharp white silhouette on fluoroscopy screens.

Bilingual Terminology -- Chemistry Discoveries and Miscellaneous

English Term	Hindi Term	Key Anchor
Alchemy (रसायन शास्त्र का प्राचीन रूप)	कीमिया / रसायन विद्या	Medieval precursor of chemistry; sought to transmute metals to gold
Lavoisier (लावोइसिये)	अन्टोइन लावोइसिये	"Father of Modern Chemistry"; law of conservation of mass; named oxygen (1774)
Dalton's Atomic Theory (डाल्टन का परमाणु सिद्धांत)	डाल्टन का परमाणु सिद्धांत	1803; atoms are indivisible; each element has unique atoms
Avogadro's Law (अवोगाद्रो का नियम)	अवोगाद्रो का नियम	Equal volumes of gases at same T and P contain equal number of molecules
Gay-Lussac's Law (गे-लुसाक का नियम)	गे-लुसाक का नियम	Gases combine in simple ratios of volumes; e.g., H2 + Cl2 = 2HCl

English Term	Hindi Term	Key Anchor
Haber Process (हैबर विधि)	हैबर प्रक्रम	N2 + 3H2 = 2NH3; conditions: 400-500 degC, 200 atm, iron catalyst; makes ammonia
Contact Process (संपर्क विधि)	सम्पर्क प्रक्रम	Industrial manufacture of H2SO4; 2SO2 + O2 = 2SO3 (V2O5 catalyst)
Solvay Process (सोल्वे विधि)	साल्वे प्रक्रम	Industrial production of Na2CO3 (soda ash/washing soda) from salt + ammonia + CO2
Ostwald Process (ऑस्टवाल्ड विधि)	ऑस्टवाल्ड प्रक्रम	Industrial production of HNO3 (nitric acid) from ammonia; Pt-Rh catalyst
Dry Ice (शुष्क बर्फ)	सूखी बर्फ	Solid CO2; -78.5 degC; sublimes directly; refrigerant, special effects
Heavy Water (भारी जल)	भारी जल	D2O (deuterium oxide); used as moderator in nuclear reactors
Laughing Gas (नाइट्रस ऑक्साइड)	नाइट्रस ऑक्साइड / हँसाने वाली गैस	N2O; anaesthetic; greenhouse gas; discovered by Joseph Priestley
Noble Gas (उत्कृष्ट गैस)	अक्रिय गैस	Group 18; completely filled electron shells; unreactive
Catalyst (उत्प्रेरक)	उत्प्रेरक	Speeds up reaction without being consumed; positive or negative
Le Chatelier's Principle (ले शातेलिए का सिद्धांत)	ले शातेलिए का नियम	System in equilibrium shifts to counter any applied change
Electrochemistry (विद्युत-रसायन)	विद्युत-रसायन	Study of relationship between electricity and chemical reactions

One-Liner Revision -- Chemistry Discoveries and Miscellaneous

#	Fast Fact
1	"Father of Modern Chemistry" = **Antoine Lavoisier** (France); named oxygen, hydrogen
2	Law of Conservation of Mass = **Lavoisier**; mass is neither created nor destroyed
3	Dalton's Atomic Theory (1803) = atoms indivisible; each element has unique identical atoms
4	Avogadro's Number = **6.022 x 10^23** entities per mole
5	Haber Process: N2 + 3H2 = 2NH3; 400-500 degC; 200 atm; Fe catalyst; makes fertiliser
6	Contact Process: SO2 + O2 = SO3 (V2O5 catalyst); SO3 + H2SO4 = oleum -> H2SO4
7	Solvay Process = produces Na2CO3 (washing soda) and NaHCO3 (baking soda) industrially
8	Ostwald Process = 4NH3 + 5O2 = 4NO + 6H2O (Pt-Rh catalyst, 850 degC); makes HNO3
9	Oxygen discovered = **Carl Scheele (1772)** and Joseph Priestley (1774); named by Lavoisier

#	Fast Fact
10	Hydrogen discovered = **Henry Cavendish (1766)**; named "inflammable air"
11	Nitrogen discovered = **Daniel Rutherford (1772)**; Lavoisier named it
12	Chlorine discovered = **Carl Wilhelm Scheele (1774)**; named by Humphry Davy
13	Dry ice = solid CO2 (-78.5 degC); sublimes (solid to gas directly); used in fog machines
14	Heavy water (D2O) = deuterium oxide; moderator in CANDU and Indian PHWR reactors
15	Laughing gas = N2O (nitrous oxide); anaesthetic; also greenhouse gas
16	Helium = lightest noble gas; non-flammable; used in balloons, MRI cooling
17	Noble gases discovered: Argon = Rayleigh + Ramsay (1894); Helium = Ramsay (1895)
18	Neon lights = neon gas in tube glows red-orange; other colours use different gases
19	Ozone (O3) = allotrope of oxygen; formed by UV + O2; both protective (stratosphere) and pollutant (troposphere)
20	Le Chatelier's Principle: equilibrium shifts against applied stress (T, P, concentration)
21	Law of mass action: rate = k[A]^m[B]^n (reactant concentrations)
22	pH of blood = 7.4; gastric acid = 1.5-3.5; milk = 6.5-6.7; lemon juice = 2-3
23	Galvanic cell = chemical energy -> electrical energy; Daniel cell = Zn-Cu
24	Electrolytic cell = electrical energy -> chemical energy; electroplating
25	Faraday's Laws of Electrolysis: 1st = mass proportional to charge; 2nd = mass proportional to equivalent weight
26	Electroplating = depositing thin layer of metal (Ag, Au, Cr, Ni) on another object
27	Anodising = electrochemical oxidation of aluminium surface; corrosion protection
28	Catalytic converter in cars = Pt-Rh-Pd catalyst; converts CO, NOx, VOCs to CO2, N2, H2O
29	Green Chemistry 12 principles = prevent waste; atom economy; safer chemicals; etc. (Paul Anastas)
30	India's 1st Chemistry Nobel (so far) = none; C.V. Raman (1930) was in Physics